Three approaches to lenses over a base

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MOTIVATION & OVERVIEW

$$\begin{tikzcd}
 & \boxed{\text{functor}} \arrow[dl, "\text{property}"'] \arrow[dr, "\text{structure}"] & \\
\boxed{\text{opfibration}} \arrow[dr, "\text{structure}"'] & & \boxed{\text{lens}} \arrow[dl, "\text{property}"] \\
 & \boxed{\begin{array}{c}\text{split}\\ \text{opfibration}\end{array}} &
\end{tikzcd}$$

1. Fibred approach

$$\boxed{\mathcal{L}ens(B) \simeq [\mathbb{H}B, s\mathbb{M}ult]_{lax}}$$

2. Algebraic approach

$$\boxed{\mathcal{L}ens(B) \xrightarrow{\text{monadic}} \mathcal{C}at/B}$$

3. Coalgebraic approach

$$\boxed{\mathcal{L}ens(B) \xrightarrow{\text{comonadic}} \mathcal{C}of(B)}$$

BACKGROUND: WHAT IS A LENS?

A **lens** is a functor equipped with a suitable choice of lifts.

$$\begin{tikzcd} A \arrow[d, "f"'] \\ B \end{tikzcd} \qquad \begin{tikzcd} a \arrow[r, "\varphi(a,u)"] \arrow[d, dotted, no head] & a' \arrow[d, dotted, no head] \\ fa \arrow[r, "u"] & b \end{tikzcd}$$

A **split opfibration** is a lens whose chosen lifts are **opcartesian**.

Proposition: A lens $A \xrightarrow{(f,\varphi)} B \simeq$ a commutative diagram of functors,

$$\begin{tikzcd} A \arrow[dd, "f"'] & \\ & X \arrow[ul, "\varphi"'] \arrow[dl, "\bar{\varphi}"] \\ B & \end{tikzcd}$$

where φ is **bijective-on-objects** and $\bar{\varphi}$ is a **discrete opfibration**.

THE CATEGORY OF LENSES OVER A BASE

For each small category B, there is a category $\mathcal{L}ens(B)$ whose:

- objects are lenses into B;
- morphisms are given by:

$$\begin{tikzcd}
A \arrow[rr, "h"] \arrow[dr, bend right, "f"'] & & C \arrow[dl, bend left, "g"] \\
X \arrow[u, "\varphi"] \arrow[rr] \arrow[dr, "\bar{\varphi}"'] & & Y \arrow[u, "\gamma"'] \arrow[dl, "\bar{\gamma}"] \\
& B &
\end{tikzcd}$$

i.e. functors $h : A \to C$ such that $gh = f$ & $h\varphi(a,u) = \gamma(ha,u)$.

- This defines a functor:

$$\mathcal{L}ens(-) : \mathrm{Cat}^{op} \longrightarrow \mathrm{CAT}$$

- There are forgetful functors:

$$\mathcal{L}ens(B) \longrightarrow \mathrm{Cat}/B$$

$$\mathcal{L}ens(B) \longrightarrow \mathrm{Cof}(B)$$

- There are full subcategories:

$$\mathrm{DOpf}(B) \overset{T}{\hookrightarrow} \mathcal{L}ens(B)$$

$$\mathrm{SOpf}(B) \hookrightarrow \mathcal{L}ens(B)$$

FIBRED APPROACH TO LENSES

Many kinds of morphism can be understood "fibre-wise":

$$\mathrm{SOpf}(B) \simeq [B, \mathrm{Cat}]$$

$$\mathrm{DOpf}(B) \simeq [B, \mathrm{Set}]$$

$$\mathrm{Cat}/B \simeq [\mathbb{H}B, \mathbb{S}\mathrm{pan}]_{\mathrm{lax}}$$

Can we find a double category $\mathbb{D}$ which classifies lenses?

$$\mathcal{L}\mathrm{ens}(B) \simeq [\mathbb{H}B, \mathbb{D}]_{\mathrm{lax}}$$

Let $\mathbb{s}/\mathbb{M}\mathrm{ult}$ be the double category of sets, functions & split multi-valued functions.

$$A \rightrightarrows X \longrightarrow B$$

There are identity-on-objects & vert. arrows double functors:

$$\mathbb{Q}\mathrm{Set} \underset{R}{\overset{L}{\underset{\longleftarrow}{\overset{\hookrightarrow}{\perp}}}} \mathbb{s}/\mathbb{M}\mathrm{ult} \longrightarrow \mathbb{S}\mathrm{pan}$$

FIBRED APPROACH TO LENSES

Theorem: There is an equivalence:

$$\mathcal{L}ens(B) \simeq [\mathbb{H}B, s\mathbb{M}ult]_{lax}$$

Idea: Lax functor $\rightsquigarrow$ lens

$$\begin{tikzcd}
\int LRF \arrow[r, "b.o.o."] \arrow[dr, "lens"] & \int F \arrow[r] \arrow[d] \arrow[dr, phantom, "\swarrow"] & \mathbb{1} \arrow[d, "*"] \\
& B \arrow[r, "LRF"] \arrow[r, bend right, "F"'] \arrow[r, phantom, shift right=3ex, "\Downarrow \varepsilon F"] & s\mathbb{M}ult
\end{tikzcd}$$

discrete opfibration

Split opfibration $\simeq$ lax functor $F : \mathbb{H}B \longrightarrow s\mathbb{M}ult$ such that

$$\begin{tikzcd}
F_x \arrow[r, "LRF(u)"] \arrow[d, equal] \arrow[dr, phantom, "\Downarrow \varepsilon_{F(u)}"] & F_y \arrow[r, "F(1_y)"] \arrow[d, equal] & F_y \arrow[d, equal] \\
F_x \arrow[r, "F(u)"] \arrow[d, equal] \arrow[dr, phantom, "\Downarrow comp(u, 1_y)"] & F_y \arrow[r, "F(1_y)"] & F_y \arrow[d, equal] \\
F_x \arrow[rr, "F(u)"'] & & F_y
\end{tikzcd}$$

is an isocell for all $u : x \longrightarrow y \in B$.

ALGEBRAIC APPROACH TO LENSES

Examples of lenses are morphisms with algebraic structure:

$$\mathrm{SOpf}(B) \xrightarrow{\text{monadic}} \mathrm{Cat}/B$$

$$\mathrm{DOpf}(B) \xrightarrow{\text{monadic}} \mathrm{Set}/B_0$$

Leads to generalisations:

- $\mathrm{Cat} \rightsquigarrow$ 2-category with pullbacks and comma objects.
- $\mathrm{Set} \rightsquigarrow$ category with pullbacks.

Johnson-Rosebrugh (2013): Lenses are certain algebras for a semi-monad:

$$\mathrm{Cat}/B \longrightarrow \mathrm{Cat}/B$$

$$A \xrightarrow{f} B \longmapsto fi_A \downarrow B \xrightarrow{\pi} B$$

Can we show that

$$\mathcal{L}\mathrm{ens}(B) \longrightarrow \mathrm{Cat}/B$$

is monadic? Can we generalise lenses by replacing Cat?

ALGEBRAIC APPROACH TO LENSES

Recall that Cat has:

- An idempotent comonad (disc. objs.).
- An O.F.S. (initial, disc. opfibration).

Theorem: $\mathcal{L}ens(B) \xrightarrow{\text{monadic}} Cat/B$

Idea:

b.o.o.

$$\begin{tikzcd}
A_0 \arrow[r, two heads] \arrow[d, "s"'] & I \arrow[d, "s"] \arrow[ddr, bend left, "\text{discrete opfibration}"] & \\
A \arrow[r] \arrow[drr, bend right, "f"'] & \bullet \arrow[dr, dashed] \arrow[ur, phantom, "\text{lens}"] & \\
 & & B
\end{tikzcd}$$

Corollary:

$$\begin{tikzcd}
SOpf(B) \arrow[rr, hook, "\perp"] \arrow[dr, shift left, "\top"] & & \mathcal{L}ens(B) \arrow[dl, shift left, "\top"] \\
 & Cat/B \arrow[ul, shift left] \arrow[ur, shift left] &
\end{tikzcd}$$

Bonus: Can replace Cat with $\mathcal{C}$, category t/w idem. comonad & O.F.S.

Example: $SpEpi(B) \xrightarrow{\text{monadic}} \mathcal{C}/B$

- The initial object comonad.
- The O.F.S. (all morphisms, iso).

COALGEBRAIC APPROACH TO LENSES

There is a category $\mathrm{Cof}(B)$ whose:

- objects are cofunctors into B;
- morphisms are commutative diagrams:

$$\begin{tikzcd}
A \arrow[r, "h"] & C \\
X \arrow[u, "\varphi"] \arrow[r] \arrow[dr, "\bar{\varphi}"'] & Y \arrow[u, "\gamma"'] \arrow[d, "\bar{\gamma}"] \\
& B
\end{tikzcd}$$

b.o.o. φ — b.o.o. γ

disc. opfibration $\bar{\varphi}$ — disc. opfibration $\bar{\gamma}$

Can we show that

$$\mathcal{L}\mathrm{ens}(B) \longrightarrow \mathrm{Cof}(B)$$

is comonadic? Generalisation?

Lemma: $\mathcal{L}\mathrm{ens}(B) \simeq \mathrm{Cof}(B)/1_B$

Idea:

$$\begin{tikzcd}
A \arrow[r, "f"] & B \arrow[d, equal] \\
X \arrow[u, "\varphi"] \arrow[r, "\bar{\varphi}"] \arrow[dr, "\bar{\varphi}"'] & B \arrow[d, equal] \\
& B
\end{tikzcd}$$

COALGEBRAIC APPROACH TO LENSES

Recall that Cat has:

- An idempotent monad (codisc. objs.).

Theorem: $\mathcal{L}ens(B) \xrightarrow{\text{comonadic}} Cof(B)$

Idea:

$$\begin{tikzcd}
X \arrow[dr, dashed, "\sim"'] \arrow[drr, bend left, "\bar{\varphi}"'] \arrow[ddr, bend right, "\varphi"'] & & \\
& \bullet \arrow[r] \arrow[d, "\wr"'] \arrow[dr, phantom, "\lrcorner", very near start] & B \arrow[d, "\wr"] \\
& A \arrow[r] & B_\infty
\end{tikzcd}$$

lens

discrete opfibration

b.o.o.

b.o.o.

codisc. cat.

Question: Is the functor $SOpf(B) \longrightarrow Cof(B)$ comonadic?

Bonus: Can replace Cat with $\mathcal{C}$, category t/w idempotent monad and class $\mathcal{M}$ of morphisms.

Example: $SpEpi(B) \xrightarrow{\text{comonadic}} B/C$

- The terminal object monad.
- $\mathcal{M}$ = class of identity morphisms

SUMMARY & FUTURE WORK

1. Fibred approach

$$\mathcal{L}ens(B) \simeq [\mathbb{H}B, s\mathbb{M}ult]_{lax}$$

2. Algebraic approach

$$\mathcal{L}ens(B) \xrightarrow{\text{monadic}} \mathcal{C}at/B$$

3. Coalgebraic approach

$$\mathcal{L}ens(B) \xrightarrow{\text{comonadic}} \mathcal{C}of(B)$$

- Understand double cat. $s\mathbb{M}ult$
- Fibred approach to $\mathcal{C}of(B)$?
- Duality between (co)algebras:

$$\mathcal{C}at/B \underset{\longrightarrow}{\overset{\longleftarrow}{\perp}} \mathcal{L}ens(B) \underset{\longrightarrow}{\overset{\longleftarrow}{\perp}} \mathcal{C}of(B)$$

- When do morphisms with algebraic structure compose?
- A formal theory of lenses

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